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Zeratul T23 软光敏调试说明

Date: Jun.2023

北京君正集成电路股份有限公司
Ingenic Semiconductor Co., Ltd.

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Release history

Date	Revision	Change
Jun.2023	1.0	First release

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1 概述

本文主要介绍在 Zeratul 平台下使用软光敏的调试方法。

1.1 软光敏的利弊

使用软光敏的利：

可节省一颗硬件光敏元器件，产品外壳无需开孔以及增加透镜，对于具有一定防水属性的产品，无需考虑开孔处的防水处理。

整体来说，可节省一定的硬件成本。

使用软光敏的弊：

由于软光敏需要依赖图像信息来判断日夜情况，需要用 2~4 帧左右用来统计 luma 信息，Sensor 配置一般是隔帧生效，所以设置新的曝光参数后需要 2~4 帧才真正生效。所以前面有 4 ~ 8 帧是无法使用的，对出图速度有一定影响。

由于软光敏是基于 ISP 的统计信息来进行日夜切换处理，且 ISP 统计信息对 Sensor 的效果参数具有一定的依赖性，所以每调试一个版本的 IQ 效果，软光敏参数也需要对应重新调试，整个软光敏调试可能会贯穿整个项目周期。

由于快起日夜判断和启动后进入应用程序的日夜判断方式有差别，所以在日夜判断的阈值调试相对麻烦，主要表现出来的问题是，快起和应用程序阈值不一致导致某些临界场景启动后出现频繁切换的问题。

综合起来，低功耗类产品建议增加硬件光敏。开发难度以及用户体验会更好。

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2 软光敏原理

2.1 原理

软光敏基于快起系统实现原理：

Sensor 固定一组曝光参数，在不同的环境光下启动并完成收敛，此时可以获得起始的平均亮度信息和收敛后的 EV 参数。把这两个参数在不同光照下进行收集，得到起始 EV 的映射表格。在建表的同时，可以使用照度计把当前环境光的照度记录下来。

在启动过程中，Sensor 同样需要固定曝光参数，在某一场景下，会获得一个起始平均亮度值（luma），通过 luma 值在映射表格中找到对应的 EV 值，然后重新配置曝光参数即可。

日夜切换也是通过 luma 值来判断，在建表的时候，已经将照度信息记录下来并与 luma 值一一对应，所以可以根据照度信息选择一个 luma 值当作日夜切换阈值即可。

软光敏应用层切换原理：

应用层切换与快起方法有差异，白天切夜视，主要依赖的画面亮度信息。夜视切白天，亮度无法作为判断标准，由于开启了红外灯，bgain 值会相对较大，所以夜视切白天，依赖亮度以及 bgain 参数作为判断依据。

建议快起和应用层日夜切换阈值可以以照度计为准，得到相同照度下的不同参数配置。防止日夜频繁切换问题。

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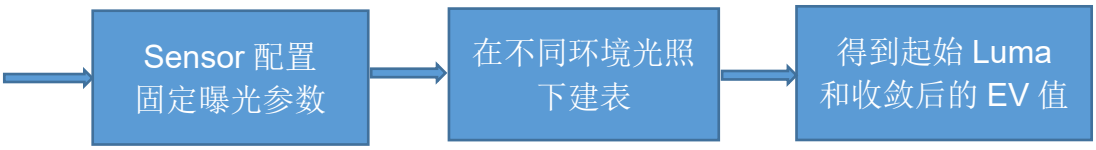
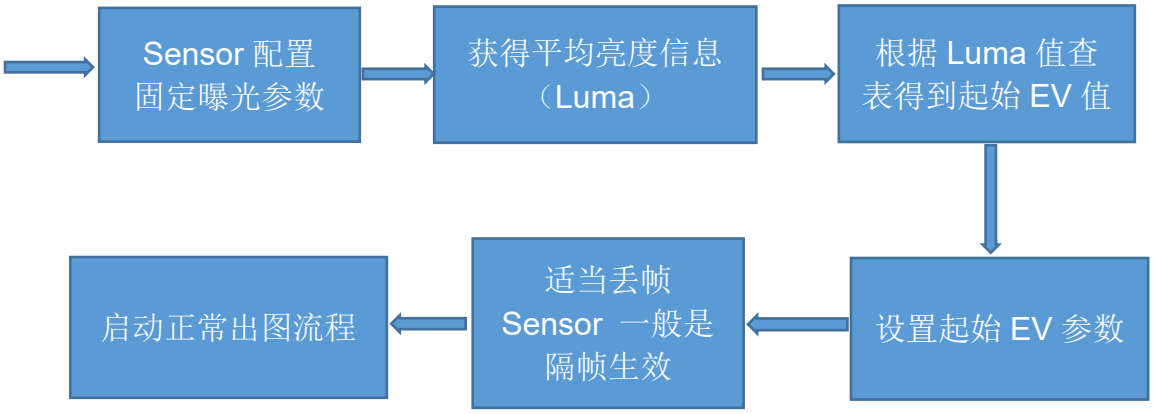
2.2 流程

建表流程：



启动流程：



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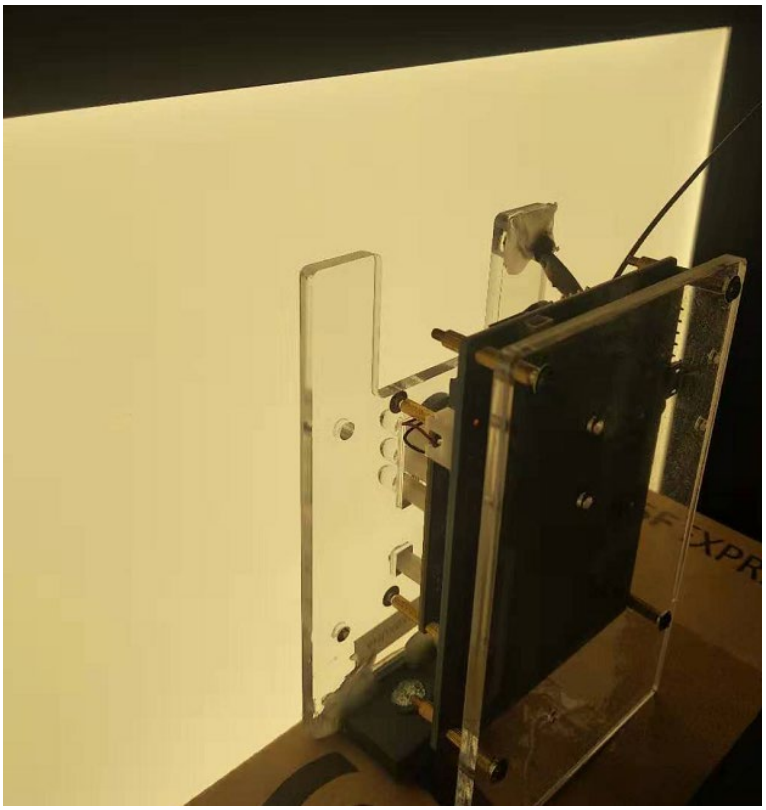
3 软光敏建表

3.1 测试环境

取一个均匀光源，放到暗室中，不要有外界光线干预，如下图放置设备，镜头可以尽量贴紧。

然后调节光源的亮度，可用照度计记录对应的照度值，从暗到亮，逐步调节，调节之后，待亮度稳定后，开关设备，通过串口读取 Luma 和 EV 值保存即可。



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3.2 曝光参数固定

起始曝光参数需要固定一下，可以参考对应的 Sensor 寄存器手册，适当修改曝光时间和增益参数。

也可以比较直接的查看 Sensor 驱动，以 jxf35 为例，找到曝光设置接口，看下实际设置曝光需要写那几个寄存器，找到寄存器对应修改即可。

```
static int jxf35_set_integration_time(struct tx_isp_subdev *sd, int value)
{
    int ret = 0;

    ret = jxf35_write(sd,  0x01, (unsigned char)(value & 0xff));
    ret += jxf35_write(sd, 0x02, (unsigned char)((value >> 8) & 0xff));
    if (ret < 0)
        pr_debug("set integration time failure!!!\n");

    return ret;
}
```

增益参数也可以使用同样方式。

```
static int jxf35_set_analog_gain(struct tx_isp_subdev *sd, int value)
{
    int ret = 0;

    ret += jxf35_write(sd, 0x00, (unsigned char)(value & 0x7f));

    if(value < 0x20) {
        ret += jxf35_write(sd, 0x2f, r2f_val | 0x20);
        ret += jxf35_write(sd, 0x0c, r0c_val | 0x40);
        ret += jxf35_write(sd, 0x80, r80_val | 0x01);
    } else {
        ret += jxf35_write(sd, 0x2f, r2f_val & 0xdf);
        ret += jxf35_write(sd, 0x0c, r0c_val & 0xbf);
        ret += jxf35_write(sd, 0x80, r80_val & 0xfe);
    }

    if (ret < 0)
        pr_debug("set analog gain!!!\n");

    return ret;
}
```

找到寄存器后，将曝光参数写入到如下文件的初始化序列当中即可。

tools/make_tag/sensor_init_setting/jxf35_t23_init.c

然后重新编译 tag 分区即可。

曝光值如何选择，建议取一个办公室白天环境下收敛之后的值即可，可以在上述 Sensor 驱动文件中增加打印来反馈该值。

另外，需要使用照度计，在记录启动 luma 值得时候，可以覆盖日夜切换区域，且可能感知的亮度范围较广即可。

注意：

初始曝光值的修改，为了防止修改出错，请找 Sensor 原厂确认。

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3.3 统计建表

打开 os/drivers/tx-isp-t23/tx-isp-fast.c 文件，将起始得 luma 值打印出来，如下：

```
771     // 获取luma信息
772     ret = tx_isp_tuning_get_ae_luma(&luma);
773     if(ret < 0) {
774         printk("tx_isp_tuning_get_ae_luma failed\n");
775         return;
776     }
777
778     // 判断日夜状态
779     g_is_night = check_is_day_or_night(luma);
780
781
782     // 切换 ISP 白天夜视模式
783     if(g_is_night == IR_STATUS_NIGHT) {
784         isp_core_tuning_switch_day_or_night(0);
785     }
786
787     // IRCUT IRLED 处理
788     irled_and_ircut_init();
789     set_irled_and_ircut(g_is_night);
790
791     // 开启sensor曝光参数设置，ISP统计结果通过I2C设置给sensor
792     tx_isp_enable_sensor_expo();
793
794     // 放开dgain的限制
795     tx_isp_tuning_set_max_dgain(196);
796
797     // 获取起始AE参数
798     g_ev = get_ae_table_value(g_start_ae_table_p, luma, g_is_night);
799     printk("%s luma:%d g_is_night:%d ev:%d\n", __func__, luma, g_is_night, g_ev);
800
```

获得luma值

打印出来

同时，打开 debug 功能，第二个参数为打印得帧数，这里建议打印多一点，比如 100 帧，debug 打开后，可以通过 dmesg 信息看到每一帧的统计信息。

```
381     }
382
383     printk("Riscv frame count:%d\n", frame_count);
384
385     g_is_night = tx_isp_get_riscv_night_mode();
386     if(g_is_night > IR_STATUS_IGNORE){
387         g_is_night = IR_STATUS_IGNORE;
388     }
389     printk("Tuning mode is:%d\n", g_is_night);
390     //g_riscv_sensor_again = 4096;
391     //g_riscv_sensor_inttime = 1242;
392
393     // Set Sensor Attr (again、inttime)
394     //tx_isp_set_ev_start(g_riscv_isp_ev);
395     //tx_isp_set_sensor_again(g_riscv_sensor_again);
396     //tx_isp_set_sensor_inttime(g_riscv_sensor_inttime);
397     //tx_isp_set_isp_gain(4096);
398
399     //tx_isp_discard_frame(0xffffff00); /* 丢弃前两帧 */
400
401     // enable debug info per frame
402     tx_isp_enable_debug(1, 50);
403
404     // 提前准备好AE Table
405     sensor_start_ae_table_init();
406
407     // 软光敏操作需要关闭前几帧sensor曝光参数设置。
408     tx_isp_disable_sensor_expo();
409
410     return 0;
411 }
```

内核统计信息打开
建议帧数可以设置大一点

修改完成后，通过 make clean && make && make release 重新编译 isp 驱动后，再重新编译内核后烧录即可。

在系统起来后，通过 dmesg 信息可以看到对应的统计信息，然后将统计信息记录到表格中就行。

```
UUID=661f7455-3453-43f6-a923-5366cc775318

Zeratul login: root
Jan  1 08:00:03 login[101]: root login on 'console'
Hello Zeratul!
[root@Zeratul:~]# dmesg
[    0.000000] Initializing cgroup subsys cpu
[    0.000000] Initializing cgroup subsys cpuacct
[    0.000000] Linux version 3.10.14-Archon (wqyan@wqyan-VirtualBox) (gcc
[    0.000000] CPU0 RESET ERROR PC:83FA065C
[    0.000000] CPU0 revision is: 00d00100 (Ingenic Xburst)
[    0.000000] FPU revision is: 00b70000
[    0.000000] cgu_get_rate, parent = 1392000000, rate = 0, m = 0, n = 0,
[    0.000000] cgu_get_rate, parent = 1392000000, rate = 0, m = 0, n = 0,
[    0.000000] CCLK:1392MHz L2CLK:696Mhz H0CLK:200MHz H2CLK:200Mhz PCLK:1(
```


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如下获得 luma 值信息：

```
[    0.158894] dma dma0chan25: Channel 25 have been requested.(phy id 6,type 0x06 desc a0a13000)
[    0.159402] dma dma0chan26: Channel 26 have been requested.(phy id 5,type 0x04 desc a0a14000)
[    0.161947] ISP [1] intt:1121 ev:1471 expr_us:33214 again:12 dgain:0 gain_log2:0 total_gain:336 AE ir
[    0.163134] TCP: cubic registered
[    0.163153] NET: Registered protocol family 17
[    0.163916] input: gpio-keys as /devices/platform/gpio-keys/input/input0
[    0.164938] Freeing unused kernel memory: 212K (804bb000 - 804f0000)
[    0.195128] ISP [2] intt:1121 ev:1471 expr_us:33214 again:10 dgain:0 gain_log2:0 total_gain:320 AE ir
[    0.226776] tx_isp_soft_photosensitive_process luma:67 g_is_night:0 ev:400
[    0.226794] TTFF frame_channel_fast_start 167 W:1920 H:1080 N:3
[    0.227572] ISP [3] intt:398 ev:1401 expr_us:11792 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt
[    0.244201] jz_sfc Build : Nov  7 2020 15:49:19
[    0.244400] the id code = c22018, the flash name is MX25L12835F
[    0.244407] the flash->board_info->quad_mode = eb
[    0.244417] JZ SFC Controller for SFC channel 0 driver register
[    0.244444] 8 cmdlinepart partitions found on MTD device jz_sfc
[    0.244450] Creating 8 MTD partitions on "jz_sfc":
[    0.244459] 0x000000000000-0x000000040000 : "boot"
[    0.255629] 0x000000040000-0x000000098000 : "tag"
```

如下获得稳定后的 EV 值信息：

```
[    2.929054] ISP [45] intt:240 ev:241 expr_us:7111 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:240 AE again:1024 AE dgain:1026 luma:65 AWB rgain:188 AWB bgain:209
[    2.995100] ISP [46] intt:238 ev:240 expr_us:7051 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:238 AE again:1024 AE dgain:1030 luma:65 AWB rgain:188 AWB bgain:209
[    3.062438] ISP [47] intt:238 ev:239 expr_us:7051 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:238 AE again:1024 AE dgain:1025 luma:64 AWB rgain:188 AWB bgain:209
[    3.128434] ISP [48] intt:236 ev:238 expr_us:6992 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:236 AE again:1024 AE dgain:1029 luma:64 AWB rgain:188 AWB bgain:209
[    3.195725] ISP [49] intt:236 ev:237 expr_us:6992 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:236 AE again:1024 AE dgain:1024 luma:64 AWB rgain:188 AWB bgain:209
[    3.261816] ISP [50] intt:234 ev:236 expr_us:6933 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:234 AE again:1024 AE dgain:1028 luma:64 AWB rgain:188 AWB bgain:209
[    3.329050] ISP [51] intt:232 ev:234 expr_us:6874 again:0 dgain:0 gain_log2:0 total_gain:258 AE intt:232 AE again:1024 AE dgain:1032 luma:63 AWB rgain:188 AWB bgain:209
[    3.395723] ISP [52] intt:232 ev:233 expr_us:6874 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:232 AE again:1024 AE dgain:1027 luma:63 AWB rgain:188 AWB bgain:209
[    3.461820] ISP [53] intt:230 ev:232 expr_us:6814 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:230 AE again:1024 AE dgain:1031 luma:63 AWB rgain:188 AWB bgain:209
[    3.529060] ISP [54] intt:230 ev:231 expr_us:6814 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:230 AE again:1024 AE dgain:1026 luma:62 AWB rgain:188 AWB bgain:209
[    3.595096] ISP [55] intt:228 ev:230 expr_us:6755 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:228 AE again:1024 AE dgain:1030 luma:62 AWB rgain:188 AWB bgain:209
[    3.662431] ISP [56] intt:228 ev:229 expr_us:6755 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:228 AE again:1024 AE dgain:1025 luma:62 AWB rgain:188 AWB bgain:209
[    3.728432] ISP [57] intt:226 ev:228 expr_us:6696 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:226 AE again:1024 AE dgain:1030 luma:62 AWB rgain:188 AWB bgain:209
[    3.795726] ISP [58] intt:226 ev:227 expr_us:6696 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:226 AE again:1024 AE dgain:1025 luma:61 AWB rgain:188 AWB bgain:209
[    3.861813] ISP [59] intt:224 ev:226 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:61 AWB rgain:188 AWB bgain:209
[    3.929043] ISP [60] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:61 AWB rgain:188 AWB bgain:209
[    3.995094] ISP [61] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:61 AWB rgain:188 AWB bgain:209
[    4.061801] ISP [62] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.128408] ISP [63] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.195078] ISP [64] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.261796] ISP [65] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.328410] ISP [66] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.395076] ISP [67] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.461785] ISP [68] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.528416] ISP [69] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.595079] ISP [70] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.661784] ISP [71] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.728408] ISP [72] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.795084] ISP [73] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.861793] ISP [74] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.928413] ISP [75] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    4.995080] ISP [76] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    5.061801] ISP [77] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    5.128409] ISP [78] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    5.195071] ISP [79] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
[    5.261791] ISP [80] intt:224 ev:225 expr_us:6637 again:0 dgain:0 gain_log2:0 total_gain:257 AE intt:224 AE again:1024 AE dgain:1030 luma:60 AWB rgain:188 AWB bgain:209
```

开始稳定

取稳定后的EV值即可

然后在不同的亮度下启动后记录对应的值即可。Luma 值范围为 0 ~ 255，所以最大统计 255 个数据即可。数据越多，EV 值设定越准确。

将建好的表格填充到以下文件对应的列表中即可：

tools/make_tag/sensor_start_ae_table/libz_jxf35.c

struct vol_start_value_table table_day[] // 白天参数列表

struct vol_start_value_table table_night[] // 夜视参数列表

然后重新编译 tag 分区后烧录即可。

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4 IRCUT 和 LRLED 操作

4.1 IRCUT、IRLED 设定

IRCUT、IRLED 等设置在 tools/make_tag/sensor_start_ae_table/ 对应的 Sensor 文件中。如下图：

```
#include "libz_common.h"

int g_gpio_led_mode = LED_MODE_HIGH_ON;
int g_gpio_led = GPIO_PB(30);
int g_gpio_ircut_mode = IRCUT_MODE_LEVEL_TRIGGER_HIGH;
int g_gpio_ircut_p = GPIO_PB(18);
int g_gpio_ircut_n = GPIO_PB(28);
int g_gpio_ircut_edge = 0;
int g_adc_value = 100;
int g_adc_direction = ADC_DIRECTION_0;
```

注意：

为了防止 uboot 也操作 IRCUT、LRLED，请将

g_gpio_led_mode = LED_MODE_IGNORE；

g_gpio_ircut_mode = IRCUT_MODE_IGNORE;

然后 ISP 驱动 tx-isp-fast.c 中根据实际情况设置初始值。

```
// AE Table Param
extern unsigned long ir_switch_mode;
extern int sensor_start_ae_table_init(void);
extern int get_sensor_start_ae_table(void);
static struct vol_start_value_table_info *g_start_ae_table_p = NULL;
static int g_gpio_led_mode = LED_MODE_HIGH_ON;
static int g_gpio_white_led_status = -1;
static int g_gpio_white_led = -1;
static int g_gpio_led = -1;
static int g_gpio_ircut_mode = IRCUT_MODE_LEVEL_TRIGGER_HIGH;
static int g_gpio_ircut_p = -1;
static int g_gpio_ircut_n = -1;
static int g_gpio_ircut_edge = -1;
static int g_adc_value = -1;
static int g_adc_reference = -1;
static int g_adc_direction = ADC_DIRECTION_0;

static ssize_t ir_fops_read(struct file *file, char __user *buf, size_t size, loff_t *ppos)
{
    unsigned int count = (unsigned int)size;
    unsigned int len = (strlen(g_ir_status[g_is_night] + *ppos) < size) ? (strlen(g_ir_status[g_is_night] + *ppos)) : size;
    ssize_t ret;

    ret = copy_to_user(buf, (void *)(g_ir_status[g_is_night] + *ppos), len);
    if(ret != 0) {
        return ret;
~/work/zeratul/os/drivers/tx-isp-t31/tx-isp-fast-soft-photosensitive.c[+][POS=129,61][15%]10/11/20 - 21:35
```

按实际情况正确设置初始值

并且获取 tag 参数的时候不进行赋值，直接使用初始值即可。

```
static int get_ae_table_info(void)
{
    struct vol_start_value_table_info *ae_start_table = get_sensor_start_ae_table();
    uint32_t ae_table_head;
    uint8_t *ae_table_head_p = (uint8_t *)&ae_table_head;
    ae_table_head_p[0] = 'A';
    ae_table_head_p[1] = 'T';
    ae_table_head_p[2] = 'A';
    ae_table_head_p[3] = 'B';
    if (ae_start_table->head == ae_table_head) {
        g_start_ae_table_p = ae_start_table;
        g_gpio_white_led_status = ae_start_table->gpio_white_led_status;
        g_gpio_white_led = ae_start_table->gpio_white_led;
        //g_gpio_led_mode = ae_start_table->gpio_led_mode;
        g_gpio_led = ae_start_table->gpio_led;
        //g_gpio_ircut_mode = ae_start_table->gpio_ircut_mode;
        g_gpio_ircut_p = ae_start_table->gpio_ircut_p;
        g_gpio_ircut_n = ae_start_table->gpio_ircut_n;
        g_gpio_ircut_edge = ae_start_table->gpio_ircut_edge;
        g_adc_value = ae_start_table->adc_value;
        g_adc_reference = ae_start_table->adc_reference;
        g_adc_direction = ae_start_table->adc_direction;
    } else {
        printk("Invalid ae table\n");
        return -1;
    }

    return 0;
}

/**
 * 获取ae table 参数
 * addr: ae table 参数地址
 * vol: ADC 采样数据
 * mode: 0 白天模式、1 夜晚模式
 */
static unsigned int get_ae_table_value(struct vol_start_value_table_info *addr, int vol, int mode)
{
~/work/zeratul/os/drivers/tx-isp-t31/tx-isp-fast-soft-photosensitive.c[+][POS=554,1][65%]10/11/20 - 21:35
```

注释不读取tag中的参数，
走驱动初始值参数

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具体参数含义如下图，对应的宏定义请参考 libz_common.h 文件说明，其中 luma 值的阈值判断是与 g_adc_value 进行对比。可以理解为 luma 值与硬光敏的 ADC 概念一致。

```
/**
 * IRLED
 * -1: 不受控制
 * 0: 高电平亮灯，低电平灭灯
 * 1: 高电平灭灯，低电平亮灯
 */
int gpio_led_mode;
int gpio_led;

/**
 * IRCUT
 * -1: 不受控制
 * 0: 电平触发方式 P 高电平 N 低电平 切夜视
 * 1: 电平触发方式 N 高电平 P 低电平 切夜视
 * 2: 沿触发方式 上升沿触发 切夜视
 * 3: 沿触发方式 下降沿触发 切夜视
 */
int gpio_ircut_mode;
int gpio_ircut_p;
int gpio_ircut_n;
int gpio_ircut_edge; /* 当触发方式为沿触发方式，当前成员参数GPIO值生效 */

// ADC 阈值
int adc_value; /* 白天夜视切换阈值,该值小于0,不受控制 */
int adc_direction; /* 0: 小于阈值为夜视 1:大于阈值为夜视 */
```

如果不想控制任何数据，对应的参数都设置为-1 即可，如下：

```
#include "libz_common.h"

int g_gpio_led_mode = LED_MODE_IGNORE;
int g_gpio_led = -1;
int g_gpio_ircut_mode = IRCUT_MODE_IGNORE;
int g_gpio_ircut_p = -1;
int g_gpio_ircut_n = -1;
int g_gpio_ircut_edge = 0;
int g_adc_value = -1;
int g_adc_direction = ADC_DIRECTION_0;
```

修改完成后重新编译 tag 分区即可。

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5 软光敏快起操作

5.1 内核软光敏操作

内核软光敏操作流程已经在 os/drivers/tx-isp-t23/tx-isp-fast.c 文件中完成，参考即可，如有优化修改，针对性修改即可。

简要说明一下必要的一些处理。

在 tx_isp_riscv_hf_prepare 函数中，需要准备好起始 EV 参数表格，并且关闭 Sensor 的曝光参数设置。由于起始 luma 值统计信息需要 2 ~ 3 帧，需要在固定 Sensor 曝光参数的基础上进行统计，所以需要关闭 Sensor 的设置，防止 ISP 重设 Sensor 参数。

```
398
399     //tx_isp_discard_frame(0xffffff00); /* 丢弃前两帧 */
400
401     // enable debug info per frame
402     //tx_isp_enable_debug(1, 100);
403
404     // 提前准备好AE Table
405     sensor_start_ae_table_init();
406
407     // 软光敏操作需要关闭前几帧Sensor曝光参数设置。
408     tx_isp_disable_sensor_expo();
409
```

在 tx_isp_riscv_hf_resize 函数中，如下位置，需要将 ISP 的 dgain 设置为 1x。由于 luma 值会受到 Sensor 的曝光、Again、ISP Dgain 影响，为了防止 ISP Dgain 在不同场景下发生编码导致 luma 值变化，所以这里将 ISP Dgain 设置为 1x。那么 luma 值影响的只与 Sensor 固定参数有关。

```
496
497     // 软光敏操作需要关闭dgain操作，防止dgain影响。
498     tx_isp_tuning_set_max_dgain(0); // 设置dgain为 1x
499
500     // 创建 ir_status 节点
501     proc_create("ir_status", S_IRUGO, NULL, &ir_status_proc_fops);
502
```

当每一帧的 Frame Done 中断触发后，都会进入 tx_isp_frame_done_int_handler 函数，对应的参数 count 是对应的帧计数，计数从 1 开始。由于 luma 值得统计稳定需要在第 2~3 帧，我们这里默认取第 3 帧的时候去做之后的软光敏流程，如下启动一个 work 来做，如果想进行 debug，可以把以下 if 0 打开，即可打印前面几帧的 luma 信息。可以适当调整 count 值来优化实际的效果。

```
815 DECLARE_WORK(soft_photosensitive_work, tx_isp_soft_photosensitive_process);
816
817 static int tx_isp_frame_done_int_handler(int count)
818 {
819 #if 0
820     // for debug
821     if(count <= 3) {
822         struct isp_core_ev_attr ev_attr;
823         unsigned char luma = 0;
824
825         tx_isp_tuning_get_ae_luma(&luma);
826         tx_isp_tuning_get_ev_attr(&ev_attr);
827
828         printk("soft photosensitive luma:%d ev:%d expr_us:%d ev_log2:%d again:%d dgain:%d gain_log2:%d\n",
829                 luma, ev_attr.ev, ev_attr.expr_us, ev_attr.ev_log2, ev_attr.again, ev_attr.dgain, ev_attr.gain_log2);
830     }
831 #endif
832
833     if(count == 3) {
834         schedule_work(&soft_photosensitive_work);
835     }
836
837     return 0;
838 }
```

在第 3 帧启动的 work 中实际的软光敏操作如下：

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```
// 获取luma信息
ret = tx_isp_tuning_get_ae_luma(&luma);
if(ret < 0) {
    printk("tx_isp_tuning_get_ae_luma failed\n");
    return;
}

// 判断日夜状态
g_is_night = check_is_day_or_night(luma);


// 切换 ISP 白天夜视模式
if(g_is_night == IR_STATUS_NIGHT) {
    isp_core_tuning_switch_day_or_night(0);
}

// IRCUT IRLED 处理
irled_and_ircut_init();
set_irled_and_ircut(g_is_night);

// 开启sensor曝光参数设置，ISP统计结果通过I2C设置给Sensor
tx_isp_enable_sensor_expo();

// 放开dgain的限制
tx_isp_tuning_set_max_dgain(196);

// 获取起始AE参数
g_ev = get_ae_table_value(g_start_ae_table_p, luma, g_is_night);
printk("%s luma:%d g_is_night:%d ev:%d\n", __func__, luma, g_is_night, g_ev);

// 设置起始EV
tx_isp_set_ev_start(g_ev);

// 配置缓存buffer
frame_channel_fast_start(g_ncu_buf.paddr + ALIGN_SIZE(g_ncu_buf.size));

// 切分辨率到15fps
tx_isp_tuning_set_fps(15, 1);

msleep(50);

// IRCUT 恢复idle状态，防止功耗损失和烧片
set_ircut_idle();
```

注意：

1、RISC-V 中会操作 Sensor 的曝光参数，会影响到 luma 值得统计，所以需要将 RISC-V 停用，将 tools/make_tag/riscv_fw/riscv_fw_sensor.bin 对应的 RISC-V 固件删除后重新编译 tag 分区即可。

2、为了软光敏快速处理，尽量将 Sensor 初始化序列配置为可用的最高帧率来处理，这样可以提高一些启动时间。在软光敏判断完成后再切到实际使用帧率即可。

3、在统计起始 luma 值过程当中，需要停掉 Sensor 曝光操作，以及设置 ISP Dgain 为 1x，在统计完成后，再将 Sensor 曝光设置恢复，并且将 ISP Dgain 恢复成实际产品需求值。

4、IRCUT 操作完成后，需要将 IRCUT 恢复成 IDLE 状态，防止功耗损失和烧毁 IRCUT 的风险。

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6 软光敏应用层操作

6.1 应用 sample

SDK 中 userland/5.4.0/imp-t23/samples/libimp-samples/sample-common.c 文件中有对应的软光敏参考 sample，微调参数即可。如下函数：

```
static int  g_soft_ps_running = 1;
void *sample_soft_photosensitive_ctrl(void *p)
{
    int i = 0;
    float gb_gain,gr_gain;
    float iso_buf;
    bool ircut_status = true;
    g_soft_ps_running = 1;
    int night_count = 0;
    int day_count = 0;
    //int day_oth_count = 0;
    //bayer域的 (g分量/b分量) 统计值
    float gb_gain_record = 200;
    float gr_gain_record = 200;
    float gb_gain_buf = 200, gr_gain_buf = 200;
    IMPISPRunningMode pmode;
    IMPISPEVAttr ExpAttr;
    IMPISPWB wb;
    IMP_ISP_Tuning_SetISPRunningMode(IMPISP_RUNNING_MODE_DAY);
    sample_SetIRCUT(1);

    while (g_soft_ps_running) {
        //获取曝光AE信息
~/work/zeratul/release/userland/4.7.2/imp-t31/samples/libimp-samples/sample-common.c[POS=1568,1][93%]10/11/20 - 11:52
```

白天切到夜视，主要依赖于平均亮度信息，适当调整该参数即可，如下：

```
    //平均亮度小于20，则切到夜视模式
    if (iso_buf >1900000) {
        night_count++;
        printf("night_count==%d\n",night_count);
        if (night_count>5) {
            IMP_ISP_Tuning_GetISPRunningMode(&pmode);
            if (pmode!=IMPISP_RUNNING_MODE_NIGHT) {
                printf("### entry night mode ###\n");
                IMP_ISP_Tuning_SetISPRunningMode(IMPISP_RUNNING_MODE_NIGHT);
                sample_SetIRCUT(0);
                ircut_status = true;
            }
            //切夜视后，取20个gb_gain的的最小值，作为切换白天的参考数值gb_gain_record ，gb_gain为bayer的G/B
            for (i=0; i<20; i++) {
                IMP_ISP_Tuning_GetWB_Statis(&wb);
                gr_gain =wb.rgain;
                gb_gain =wb.bgain;
                if (i==0) {
                    gb_gain_buf = gb_gain;
                    gr_gain_buf = gr_gain;
                }
                gb_gain_buf = ((gb_gain_buf>gb_gain)?gb_gain:gb_gain_buf);
                gr_gain_buf = ((gr_gain_buf>gr_gain)?gr_gain:gr_gain_buf);
                usleep(300000);
                gb_gain_record = gb_gain_buf;
                gr_gain_record = gr_gain_buf;
                // printf("gb_gain == %f,iso_buf=%f",gb_gain,iso_buf);
                // printf("gr_gain_record == %f\n ",gr_gain_record);
            }
        }
    } else {
        night_count = 0;
    }
```

夜视切白天，除了亮度信息，dgain 参数也需要参考，为了防止日夜来回切换，适当增加防抖处理。如下：

```
//满足这3个条件进入白天切换判断条件
if (((int)iso_buf < 479832) &&(ircut_status == true) &&(gb_gain>gb_gain_record+15)) {
    if ((iso_buf<361880)||(gb_gain >145)) {
        day_count++;
    } else {
        day_count=0;
    }
    // printf("gr_gain_record == %f gr_gain =%f line=%d\n",gr_gain_record,gr_gain,__LINE__);
    // printf("day_count == %d\n",day_count);
    if (day_count>3) {
        printf("### entry day mode ###\n");
        IMP_ISP_Tuning_GetISPRunningMode(&pmode);
        if (pmode!=IMPISP_RUNNING_MODE_DAY) {
            IMP_ISP_Tuning_SetISPRunningMode(IMPISP_RUNNING_MODE_DAY);
            sample_SetIRCUT(1);
            ircut_status = false;
        }
    }
} else {
    day_count = 0;
}
sleep(1);
```


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7 丢帧说明

由于 luma 值统计需要消耗几帧，并且重新设置曝光参数后，Sensor 一般是隔帧生效，实际测试不同的 Sensor 隔帧数量不一样，平均在 2~4 帧左右，所以这几帧图像数据是不能用的，需要做丢帧处理，君正平台提供了一个 tx_isp_discard_frame API 用来做丢帧处理，需要在 tx_isp_riscv_hf_prepare 函数中调用，如下：

```
330     printk("Tuning mode is:%d\n", g_is_night);
331     //g_riscv_sensor_again = 4096;
332     //g_riscv_sensor_inttime = 1242;
333
334     // Set Sensor Attr (again、inttime)
335     //tx_isp_set_ev_start(g_riscv_isp_ev);
336     //tx_isp_set_sensor_again(g_riscv_sensor_again);
337     //tx_isp_set_sensor_inttime(g_riscv_sensor_inttime);
338     //tx_isp_set_isp_gain(4096);
339
340     tx_isp_discard_frame(0xffffff00);
341
342     // enable debug info per frame
343     tx_isp_enable_debug(1, 50);
344
345     return 0;
346 }
```

注意：

丢帧 API 使用方法请参考 tx-isp-device.h 头文件描述，丢帧的起点是 Sensor 第一帧开始，和 tx_isp_frame_done_int_handler 对应的 count 值一致。

如下，丢 4 帧效果，第 5 帧开始保存图像数据。

```
[     0.249416] Creating 8 MTD partitions on "jz_sfc":
[     0.249425] 0x000000000000-0x000000040000 : "boot"
[     0.258148] 0x000000040000-0x000000098000 : "tag"
[     0.268292] 0x000000098000-0x000000598000 : "kernel"
[     0.278438] 0x000000598000-0x000000898000 : "rootfs"
[     0.288496] TTFF frame_channel_fast_start 108 W:1920 H:1080 N:3
[     0.289280] ISP [4] intt:1328 ev:1344 expr_us:39348 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:1328
[     0.290854] 0x000000898000-0x000000b18000 : "recovery"
[     0.299403] 0x000000b18000-0x000000f80000 : "system"
[     0.307837] 0x000000f80000-0x000001000000 : "config"
[     0.316366] 0x000000000000-0x000001000000 : "all"
[     0.325329] SPI NOR MTD LOAD OK
[     0.337651] jffs2: version 2.2. (NAND) (SUMMARY)  © 2001-2006 Red Hat, Inc.
[     0.352264] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[     0.355170] --- setting intergration value:1328
[     0.355828] TTFF frame_channel_buffer_done:257 Chn:0 Buf:0 Write Done:526
[     0.356380] ISP [5] intt:1312 ev:1328 expr_us:38874 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:1312
[     0.366504] zram0: detected capacity change from 0 to 16777216
[     0.371220] Adding 16380k swap on /dev/zram0.  Priority:-1 extents:1 across:16380k SS
[     0.383542] Audio codec_set_buildin_mic
[     0.401779] codec_fast_start done
[     0.421861] --- setting intergration value:1312
[     0.422528] TTFF frame_channel_buffer_done:257 Chn:0 Buf:1 Write Done:593
[     0.423137] ISP [6] intt:1296 ev:1312 expr_us:38400 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:1296
[     0.488466] --- setting intergration value:1296
[     0.489121] TTFF frame_channel_buffer_done:257 Chn:0 Buf:2 Write Done:659
[     0.489684] ISP [7] intt:1102 ev:1297 expr_us:32651 again:0 dgain:0 gain_log2:0 total_gain:256 AE intt:1102
```

如下，丢 8 帧效果，第 9 帧开始保存图像数据。按上面描述，luma 值统计以及 sensor 隔帧生效大约需要 4 ~ 8 帧，推荐丢 8 帧，也可以根据实际调试效果丢帧。

```
[     0.248779] 8 cmdlinepart partitions found on MTD device jz_sfc
[     0.248784] Creating 8 MTD partitions on "jz_sfc":
[     0.248794] 0x000000000000-0x000000040000 : "boot"
[     0.257494] 0x000000040000-0x000000098000 : "tag"
[     0.267703] 0x000000098000-0x000000598000 : "kernel"
[     0.277836] 0x000000598000-0x000000898000 : "rootfs"
[     0.287918] 0x000000898000-0x000000b18000 : "recovery"
[     0.288508] TTFF frame_channel_fast_start 108 W:1920 H:1080 N:3
[     0.289287] ISP [4] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.298841] 0x000000b18000-0x000000f80000 : "system"
[     0.307362] 0x000000f80000-0x000001000000 : "config"
[     0.315826] 0x000000000000-0x000001000000 : "all"
[     0.324703] SPI NOR MTD LOAD OK
[     0.335870] jffs2: version 2.2. (NAND) (SUMMARY)  © 2001-2006 Red Hat, Inc.
[     0.348805] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[     0.356836] ISP [5] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.362598] zram0: detected capacity change from 0 to 16777216
[     0.367322] Adding 16380k swap on /dev/zram0.  Priority:-1 extents:1 across:16380k SS
[     0.383539] Audio codec_set_buildin_mic
[     0.401880] codec_fast_start done
[     0.422367] ISP [6] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.489658] ISP [7] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.556227] ISP [8] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.621907] TTFF frame_channel_buffer_done:257 Chn:0 Buf:0 Write Done:792
[     0.623615] ISP [9] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE i
[     0.688576] TTFF frame_channel_buffer_done:257 Chn:0 Buf:1 Write Done:859
[     0.688689] ISP [10] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
[     0.755243] TTFF frame_channel_buffer_done:257 Chn:0 Buf:2 Write Done:926
[     0.756953] ISP [11] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
[     0.823500] ISP [12] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
[     0.890169] ISP [13] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
[     0.956841] ISP [14] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
[     1.023500] ISP [15] intt:1462 ev:1462 expr_us:43318 again:0 dgain:0 gain_log2:0 total_gain:256 AE
```


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8 PATCH 合并说明

该 patch 中包含以下文件：

tx-isp-t31-20201110.7z	2020/11/10 12:16	7Z 文件	415 KB
tx-isp-t31-device.c	2020/11/10 12:19	C 文件	9 KB

将 tx-isp-t23-device.c 文件替换 os/kernel/drivers/media/platform/tx-isp-t23 目录下的文件即可。

将 tx-isp-t23 驱动替换 os/drivers/tx-isp-t23 目录即可，该文件中包含两个驱动文件：

tx-isp-fast.c // 原始 isp 驱动代码文件

tx-isp-fast-soft-photosensitive.c // 软光敏流程 isp 驱动代码文件

手动将 tx-isp-fast-soft-photosensitive.c 重命名为 tx-isp-fast.c 即可：

```
mv tx-isp-fast-soft-photosensitive.c tx-isp-fast.c
```

然后重新 make clean && make && make release 重新编译 isp 驱动，再重新编译内核后烧录即可。

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